

Government College Women University, Faisalabad (GCWUF)

Department of Computer Science

ASSIGNMENT

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Roll No	2506
Course Title	Software Engineering
Course Code	CSC-415
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1. Project Title

Smart Grocery Expiry and Inventory Tracker

The Smart Grocery Expiry and Inventory Tracker is a software system that helps grocery stores and households record their grocery items, keep track of stock quantity, monitor expiry dates and receive timely alerts before products expire.

2. Need of the Project

Groceries such as milk, bread, juices, canned food, spices and frozen items have limited shelf life. Today, most small and medium grocery stores and many homes still manage stock manually using registers, memory or simple notes. This creates many problems:

- **Expired products:** Products expire on the shelf or in the kitchen without anyone noticing, which causes financial loss and food waste.
- **Inaccurate inventory:** Manual stock counting takes time, and mistakes in records are common.
- **Overstocking and shortage:** Items run out of stock because the shortage is noticed too late, or too much stock is bought and wasted.
- **Health and trust issues:** Selling an expired product can harm customers and damage the reputation of the store.
- **No reports or planning:** There is no easy way to know which items are selling fast, which are slow, and what should be ordered next.

Because of these problems, a smart, simple and automated system is needed to manage inventory and expiry dates efficiently.

3. Proposed Solution

The proposed solution is a software application (mobile/web based) that stores all grocery items with their quantity, purchase date and expiry date. The system automatically checks the expiry dates and sends alerts, so that users can sell or use the products on time. It solves the problems listed above in the following way:

3.1 Main Features

- **Inventory management:** Add, update and delete grocery items with name, category, quantity, price and expiry date.
- **Easy item entry:** Items are added by manual entry or by scanning a barcode for faster work.
- **Expiry alerts:** Notifications are sent for items that are about to expire (for example 7, 3 and 1 day before) and for items already expired.
- **Low-stock alerts:** Warning is shown when the quantity of an item falls below a set minimum level.
- **Smart sorting:** Items are sorted by nearest expiry date so that older stock is sold first (FEFO – First Expired, First Out).
- **Reports:** Reports on current stock, expired items, near-expiry items and wasted products.
- **Shopping list:** The home user can create a shopping list from low or finished items.

3.2 How the Problem Is Solved

- **Problem – expired products** → Expiry alerts reduce the loss caused by expired products.
- **Problem – wrong inventory** → Real-time digital records replace manual counting.
- **Problem – overstock/shortage** → Low-stock alerts and reports guide correct ordering.
- **Problem – health risk** → Expired items are flagged so they are not sold or used.
- **Problem – no planning** → Reports show fast-selling and slow-selling products for better planning.

4. Objectives of the Project

4.1 General Objective

To develop a user-friendly Smart Grocery Expiry and Inventory Tracker that reduces food waste and financial loss by managing grocery stock and expiry dates efficiently.

4.2 Specific Objectives

- To maintain a digital record of all grocery items with quantity and expiry date.
- To send automatic alerts for near-expiry, expired and low-stock items.
- To reduce the number of expired products sold or wasted.
- To save time and effort by replacing manual stock checking.
- To provide reports that help owners in decision making and purchase planning.
- To provide a simple interface that can be used easily by store staff and home users.
- To improve food safety and customer trust.

5. Stakeholders

Stakeholders are the people, groups or organizations who are affected by the system or who affect the system. The main stakeholders of this project and their complete information are given below.

Stakeholder	Role / Description	Interest and Needs	Expectations
Grocery Store Owner / Manager	Owns the store and manages stock, purchasing and sales.	Wants to reduce expired-stock loss, track inventory in real time and see reports.	Accurate stock data, expiry alerts, reports and higher profit.
Store Staff / Stock Keeper	Enters items, updates stock and arranges shelves.	Needs an easy system to add items, scan or enter expiry dates and update quantity.	Simple interface, fast data entry, less manual work.
Household Customer (Home User)	Buys groceries and stores them at home.	Wants to avoid wasting food and money by tracking expiry dates at home.	Reminders before expiry, shopping list and easy mobile use.
Suppliers / Distributors	Provide products to the store.	Need to know current stock levels and demand for timely supply.	Clear reorder requests and less returned or expired goods.
Project Developer / Team	Analyses, designs, develops and tests the system.	Wants clear requirements and a successful, working project.	Clear requirements, feedback and timely completion.

Stakeholder	Role / Description	Interest and Needs	Expectations
Project Supervisor (Mam Sadia Saher)	Guides and evaluates the project.	Wants the project to follow software engineering principles.	Proper documentation, design and quality work.
Food Safety Authority / Government	Regulates food quality and safety.	Wants expired products not to be sold to the public.	Compliance with food safety rules and public health.

5.1 Types of Stakeholders

- **Primary (direct) stakeholders:** Store owner/manager, store staff and home customers, who directly use the system.
- **Secondary stakeholders:** Suppliers, project developer/team and supervisor, who are involved in or support the system.
- **Tertiary (indirect) stakeholders:** Food safety authority/government and the general public, who are indirectly affected by the system.